

OUTPUT:

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*** Initializing Emergency Patient System ***
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*** Adding Initial Patients ***
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Patient P001 added successfully  
Patient P002 added successfully  
Patient P003 added successfully  
Patient P004 added successfully  
Patient P005 added successfully  
Patient P006 added successfully  
Patient P007 added successfully
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```
*** Displaying All Waiting Patients ***
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--- Current Waiting Patients Queue (Severity, then Arrival Order) ---
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P002 | Severity: 1 | Arrival: 2 | Doctor: Dr. Priya  
P004 | Severity: 1 | Arrival: 4 | Doctor: Dr. Meena  
P003 | Severity: 2 | Arrival: 3 | Doctor: Dr. Kumar  
P006 | Severity: 2 | Arrival: 6 | Doctor: Dr. Patel  
P001 | Severity: 3 | Arrival: 1 | Doctor: Dr. Arun  
P005 | Severity: 4 | Arrival: 5 | Doctor: Dr. Singh  
P007 | Severity: 5 | Arrival: 7 | Doctor: Dr. Gupta
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*** Next Patient in Queue ***
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Next patient in queue: P002 | Severity: 1 | Arrival: 2 | Doctor: Dr. Priya
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*** Treating the Next Patient ***
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Patient P002 sent for treatment: P002 | Severity: 1 | Arrival: 2 | Doctor: Dr. Priya
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```
*** Displaying Queue After One Treatment ***
--- Current Waiting Patients Queue (Severity, then Arrival Order) ---
P004 | Severity: 1 | Arrival: 4 | Doctor: Dr. Meena
P003 | Severity: 2 | Arrival: 3 | Doctor: Dr. Kumar
P006 | Severity: 2 | Arrival: 6 | Doctor: Dr. Patel
P001 | Severity: 3 | Arrival: 1 | Doctor: Dr. Arun
P005 | Severity: 4 | Arrival: 5 | Doctor: Dr. Singh
P007 | Severity: 5 | Arrival: 7 | Doctor: Dr. Gupta
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*** Adding More Patients ***
Patient P008 added successfully
Patient P009 added successfully
Patient P010 added successfully

*** Displaying Queue After Adding More Patients ***
--- Current Waiting Patients Queue (Severity, then Arrival Order) ---
P004 | Severity: 1 | Arrival: 4 | Doctor: Dr. Meena
P008 | Severity: 1 | Arrival: 8 | Doctor: Dr. Priya
P003 | Severity: 2 | Arrival: 3 | Doctor: Dr. Kumar
P006 | Severity: 2 | Arrival: 6 | Doctor: Dr. Patel
P010 | Severity: 2 | Arrival: 10 | Doctor: Dr. Singh
P001 | Severity: 3 | Arrival: 1 | Doctor: Dr. Arun
P009 | Severity: 3 | Arrival: 9 | Doctor: Dr. Kumar
P005 | Severity: 4 | Arrival: 5 | Doctor: Dr. Singh
P007 | Severity: 5 | Arrival: 7 | Doctor: Dr. Gupta
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*** Searching for Patient P003 ***

Found patient: P003 | Severity: 2 | Arrival: 3 | Doctor: Dr. Kumar

*** Updating Patient P003: Severity to 1, Doctor to Dr. Ravi ***

Patient P003 updated successfully and queue re-prioritized

*** Displaying Queue After Updating P003 ***

--- Current Waiting Patients Queue (Severity, then Arrival Order) ---

P003	Severity: 1	Arrival: 3	Doctor: Dr. Ravi
P004	Severity: 1	Arrival: 4	Doctor: Dr. Meena
P008	Severity: 1	Arrival: 8	Doctor: Dr. Priya
P006	Severity: 2	Arrival: 6	Doctor: Dr. Patel
P010	Severity: 2	Arrival: 10	Doctor: Dr. Singh
P001	Severity: 3	Arrival: 1	Doctor: Dr. Arun
P009	Severity: 3	Arrival: 9	Doctor: Dr. Kumar
P005	Severity: 4	Arrival: 5	Doctor: Dr. Singh
P007	Severity: 5	Arrival: 7	Doctor: Dr. Gupta

*** Treating Two More Patients ***

Patient P003 sent for treatment: P003 | Severity: 1 | Arrival: 3 | Doctor: Dr. Ravi

Patient P004 sent for treatment: P004 | Severity: 1 | Arrival: 4 | Doctor: Dr. Meena

*** Displaying Queue After More Treatments ***

--- Current Waiting Patients Queue (Severity, then Arrival Order) ---

P008	Severity: 1	Arrival: 8	Doctor: Dr. Priya
P006	Severity: 2	Arrival: 6	Doctor: Dr. Patel
P010	Severity: 2	Arrival: 10	Doctor: Dr. Singh
P001	Severity: 3	Arrival: 1	Doctor: Dr. Arun
P009	Severity: 3	Arrival: 9	Doctor: Dr. Kumar
P005	Severity: 4	Arrival: 5	Doctor: Dr. Singh
P007	Severity: 5	Arrival: 7	Doctor: Dr. Gupta

*** Searching for a Non-existent Patient (P999) ***

Patient P999 not found

*** Updating an Already Treated Patient (P002) ***

Patient P002 updated successfully

*** Displaying Final Queue ***

--- Current Waiting Patients Queue (Severity, then Arrival Order) ---

P008	Severity: 1	Arrival: 8	Doctor: Dr. Priya
P006	Severity: 2	Arrival: 6	Doctor: Dr. Patel
P010	Severity: 2	Arrival: 10	Doctor: Dr. Singh
P001	Severity: 3	Arrival: 1	Doctor: Dr. Arun
P009	Severity: 3	Arrival: 9	Doctor: Dr. Kumar
P005	Severity: 4	Arrival: 5	Doctor: Dr. Singh
P007	Severity: 5	Arrival: 7	Doctor: Dr. Gupta

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*** Attempting to treat from an empty queue (should print 'No patients waiting') ***
Patient P008 sent for treatment: P008 | Severity: 1 | Arrival: 8 | Doctor: Dr. Priya
Patient P006 sent for treatment: P006 | Severity: 2 | Arrival: 6 | Doctor: Dr. Patel
Patient P010 sent for treatment: P010 | Severity: 2 | Arrival: 10 | Doctor: Dr. Singh
Patient P001 sent for treatment: P001 | Severity: 3 | Arrival: 1 | Doctor: Dr. Arun
Patient P009 sent for treatment: P009 | Severity: 3 | Arrival: 9 | Doctor: Dr. Kumar
Patient P005 sent for treatment: P005 | Severity: 4 | Arrival: 5 | Doctor: Dr. Singh
Patient P007 sent for treatment: P007 | Severity: 5 | Arrival: 7 | Doctor: Dr. Gupta
No patients waiting
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*** Final Patient List (including treated ones, for verification) ***
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P001	Status: Treated	Severity: 3	Doctor: Dr. Arun
P002	Status: Treated	Severity: 5	Doctor: Dr. House
P003	Status: Treated	Severity: 1	Doctor: Dr. Ravi
P004	Status: Treated	Severity: 1	Doctor: Dr. Meena
P005	Status: Treated	Severity: 4	Doctor: Dr. Singh
P006	Status: Treated	Severity: 2	Doctor: Dr. Patel
P007	Status: Treated	Severity: 5	Doctor: Dr. Gupta
P008	Status: Treated	Severity: 1	Doctor: Dr. Priya
P009	Status: Treated	Severity: 3	Doctor: Dr. Kumar
P010	Status: Treated	Severity: 2	Doctor: Dr. Singh